

Securing the Internet of Medical Things: An Overview of AI–Blockchain–Based Approaches

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Index Terms—IoMT, AI, blockchain, smart health system

I. INTRODUCTION

The digital transformation of healthcare has accelerated rapidly in recent years, driven by the widespread adoption of connected medical devices in wearable sensors and intelligent data analytics [1]. This evolution is commonly referred to as the Internet of Medical Thing (IoMT), a paradigm in which medical sensors, wearable devices and clinical systems are interconnected to enable continuous monitoring, remote diagnostics and data-driven healthcare services [4], [7]. IoMT-based solutions promise substantial benefits, including improved quality of care, reduced healthcare costs and enhanced accessibility to healthcare services, particularly for patients with chronic conditions in remote settings [3], [4]. Recent developments in the field of artificial intelligence (AI) have led to significant advancements in the context of IoMT systems, with the capacity to transform raw medical data into valuable insights. The application of machine learning (ML) and deep learning (DL) techniques has seen a considerable rise in the fields of disease prediction, anomaly detection and personalised treatment recommendations [3], [4]. Surveys on AI-enabled healthcare systems highlight that AI significantly enhances the value of IoMT by enabling real-time decision-making and predictive analytics on large-scale physiological data streams [8]. However, the integration of AI also amplifies existing security and privacy concerns, as sensitive health data must be collected, transmitted, processed and stored across heterogeneous and resource-constrained devices. Security and privacy challenges have consistently been identified as major barriers to the large-scale deployment of smart healthcare systems. Medical IoT devices often suffer from limited computational capabilities, lack of standardized security mechanisms and long device lifecycles, making them attractive targets for cyberattacks [1], [3]. Prior work on smart health wearables has shown that users are particularly concerned about unauthorized data access, secondary data usage and lack of transparency regarding how their health information is processed and shared [2], [7]. These concerns are further intensified in AI-enhanced IoMT systems, where automated

decision-making and complex data flows reduce user visibility and control. Consequently, ensuring confidentiality, integrity and availability of medical data is fundamental requirement for trustworthy IoMT deployments [2].

Beyond technical vulnerabilities, user trust and privacy perceptions play a decisive role in the acceptance of smart healthcare technologies [9]. Studies on health wearables show that users are particularly concerned about unauthorized data access, secondary data storage and lack of transparency regarding how their personal health information is processed and shared [1], [3].

Traditional centralized security architectures, which rely on cloud-based processing and trusted third parties, face significant limitations in addressing these challenges [7]. Centralised designs often introduce single points of failure, increased latency and limited audibility which are especially problematic for time-critical and safety-sensitive explored approaches to enhance trust, resilience and transparency in smart healthcare environments [2], [7], [8].

Among these approaches, blockchain technology has gained considerable attention as a means to strengthen security and data governance in IoMT systems [2], [3]. While blockchain can ensure data integrity and access control, AI can support intelligent security monitoring, anomaly detection and predictive threat mitigation [2], [6], [8]. Recent frameworks explicitly combine AI-driven security mechanisms with blockchain-based verification to address key IoMT challenges related to trust, transparency and system resilience [5], [6].

- A. *Motivation: Increasing adoption of IoMT in modern health-care*
- B. *Limitations of traditional IoT security models in medical environments*
- C. *Emerging role of AI and blockchain in improving security and trust*
- D. *Structure of the paper*

This seminar paper provides an overview of security and privacy challenges in AI-enhanced IoMT systems and examines how blockchain-based approaches can address these challenges. It surveys existing research and discusses a proposed architecture that illustrate the use of distributed ledgers to secure AI-enabled IoMT environments. By combining a broad overview of challenges with an in-depth discussion of blockchain-based solution, this work aims to contribute to a

clearer understanding of current practices and open issues in securing next-generation smart healthcare systems.

II. BACKGROUND ON AI-ENHANCED IOMT

The IoMT represents a specialized subset of the Internet of Things (IoT), focusing on healthcare-related devices and systems such as wearable sensors or smart monitoring equipment. These components are interconnected to enable continuous health data collection and real-time communication between patients, healthcare providers and medical infrastructures [3], [4]. Typical IoMT architectures consists of multiple layers, including sensing devices, communication gateways, processing units and backend healthcare services, each of which plays a critical role in the delivery of smart healthcare applications [3].

Data generated by IoMT devices is often heterogeneous in nature. It ranges from physiological signals (e.g. heart rate) to medical images and contextual patient information. To handle this complexity, IoMT systems increasingly rely on AI techniques to extract meaningful insights from large-scale and continuous data streams [3], [4]. ML and DL models are widely used for tasks, such as activity recognition, to enable more proactive and personalized healthcare services [4].

AI-enhanced IoMT systems commonly adopt layered computing architectures that distribute processing across edge, fog and cloud environments [7]. Edge and fog computing nodes are often used to perform latency-sensitive tasks close to the data source, while cloud infrastructures provide scalable resources for model training and long-term data analysis [7], [8]. This hierarchical processing model helps reduce communication delays and bandwidth consumption but also introduces additional complexity in terms of system coordination and security management [3], [7].

From a security and privacy perspective, IoMT systems impose particularly strong requirements [8], [9]. Unauthorized access, data tampering or loss of availability can have serious consequences for patient safety and trust in healthcare services [2], [4], [7]. As a result, AI-enhanced IoMT systems must ensure secure data transmission, reliable authentication and controlled access to medical information across all architectural layers. In addition to technical problems, user trust plays an important role in the adoption of IoMT technologies [8]. Studies on health wearables have shown that users are often concerned about how their data is collected, processed and shared, particularly when automated decision-making systems are involved [1], [4]. In AI-driven IoMT environments, these concerns are amplified by the opacity of ML models and the difficulty for users to understand how health-related decisions are derived from their data.

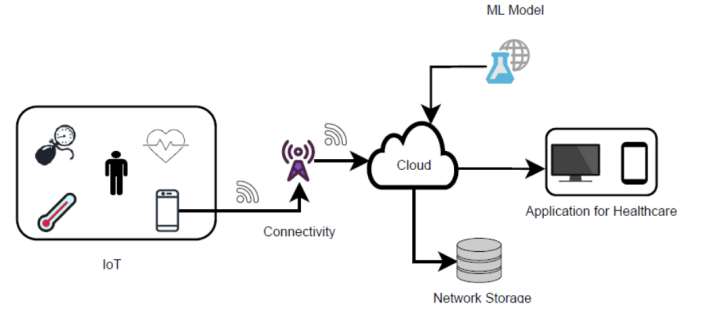
Recent research has therefor emphasized the need for more transparent and trustworthy system designs [4]. Surveys of AI-enabled healthcare systems highlight that traditional centralized approaches struggle to meet these requirements due to limited transparency, lack audibility and dependence on trusted third parties [2], [3], [8]. These limitations have motivated the exploration of decentralized technologies, such as blockchain,

to support secure data management and verifiable system behaviour in IoMT environments [5], [8].

Overall, AI-enhanced IoMT systems combine advanced sensing, intelligent data analysis and distributed computing to enable these healthcare services. At the same time, they introduce complex security, privacy and trust challenges that must be addressed holistically. This background provides the foundation for the following section, which surveys the key security and privacy challenges in AI-enhanced IoMT systems in more detail.

A. Internet of Medical Things (IoMT)

B. AI in IoMT Systems



C. Architectural characteristics of AI-Enhanced IoMT

D. Blockchain Technology for IoMT (Conceptual Overview)

III. SECURITY AND PRIVACY CHALLENGES OF AI-ENHANCED IOMT

A. Device-Level Security Challenges

B. Network and Communication Threats

C. AI-Specific Security Risks

D. Privacy Challenges

E. Trust and System-Level Issues

IV. MOTIVATION FOR COMBINING AI AND BLOCKCHAIN

A. Use of Distributed Ledgers in IoMT

AI-enhanced IoMT systems rely on continuous data exchange between medical devices, processing nodes and healthcare service providers. This data is highly sensitive and often shared across organizational and administrative boundaries, which makes trust management and secure models struggle to provide sufficient transparency and integrity in a complex and distributed healthcare environments [2], [3], [7].

Distributed ledger technologies, commonly referred to as blockchain, have therefore been proposed as a promising mechanism to strengthen security and trust in IoMT systems [5]. A blockchain maintains a shared, append-only ledger that is replicated across multiple nodes, ensuring that stored data cannot be altered without detection.

From a security perspective, blockchains can reduce reliance on trusted third parties by enabling decentralized trust establishment [5]. Instead of delegating data control to a single cloud provider, participating entities collectively maintain and validate the ledger state. This decentralization can improve system resilience and reduce single points of failure, which

are known limitations of cloud-centric healthcare architectures [5], [7].

In addition, blockchain-based systems support transparent auditing mechanisms. All transactions recorded on the ledger can be verified and traced, enabling accountability and post-event analysis. This is especially relevant in healthcare scenarios where compliance, traceability and trust between stakeholders are essential [3], [5].

Studies further emphasizes that blockchain-system IoMT systems often combine on-chain and off-chain storage strategies.

B. 4.1 Motivation for combining AI and blockchain

C. 4.2 Architectural principles

D. Blockchain-Enabled IoMT

Almalki et al. propose a blockchain-enabled IoMT framework designed to enhance security in AI-driven healthcare applications [5]. The framework integrates IoMT devices, AI components and a permissioned blockchain infrastructure to address key limitations of centralized healthcare systems, particularly those related to data tampering and reliance on trusted third parties.

1) *System Architecture Overview:* The proposed architecture follows a layered design that combines IoMT sensing devices, intermediate processing layers and a distributed ledger network. IoMT devices are responsible for collecting patient data. This data is transmitted to processing units that perform AI-based analysis. Instead of storing sensitive medical data solely in centralized databases, the framework introduces a blockchain layer to record and manage critical information related to data access, processing results and system events [5].

The blockchain component is implemented using a permissioned ledger, which restricts participation to authorized healthcare stakeholders. This design choice reflects the controlled and regulated nature of healthcare environments, where only trusted entities should be allowed to validate and access records. By distributing the ledger across multiple nodes, the system avoids single points of failure and increases resilience against data manipulation attacks [5].

2) *Role of Artificial Intelligence:* Within the framework, AI is used to analyze medical data generated by IoMT devices. Almalki et al. [5] focus on AI-driven diagnostic tasks, where ML models process health data to support clinical decision-making. The correctness and reliability of these AI outputs depend heavily on the integrity of the input data, which motivates the tight integration between AI components and the blockchain layer. By ensuring that data used for AI inference is verifiable and tamper-resistant, the blockchain helps preserve the trustworthiness of AI-generated results. Diagnostic outcomes and relevant metadata can be logged on the ledger, enabling later verification.

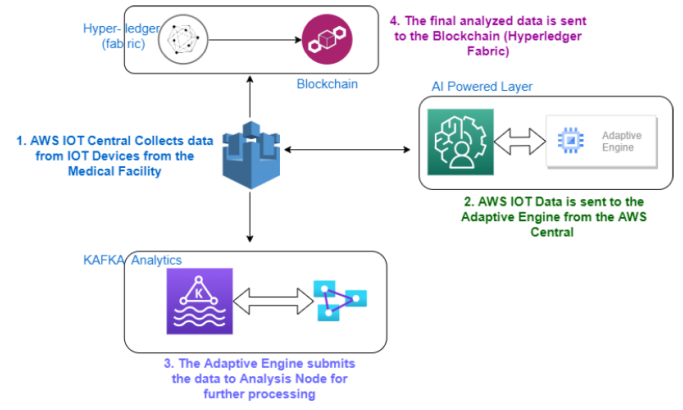


Fig. 1. Proposed System Design for the IoMT based Blockchain in Healthcare, Almalki et al. [5]

3) *Blockchain Integration and Ledger Functionality:* The blockchain layer serves as a trusted record-keeping mechanism within the framework. Rather than storing raw medical data directly on the ledger, which would be inefficient and raise privacy concerns, the framework records hashes, references and access-related metadata on the blockchain. This design enables integrity verification while keeping sensitive health data off-chain [5].

E. (AIBPSF-IoMT Security Framework)

V. DISCUSSION

VI. CONCLUSION

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